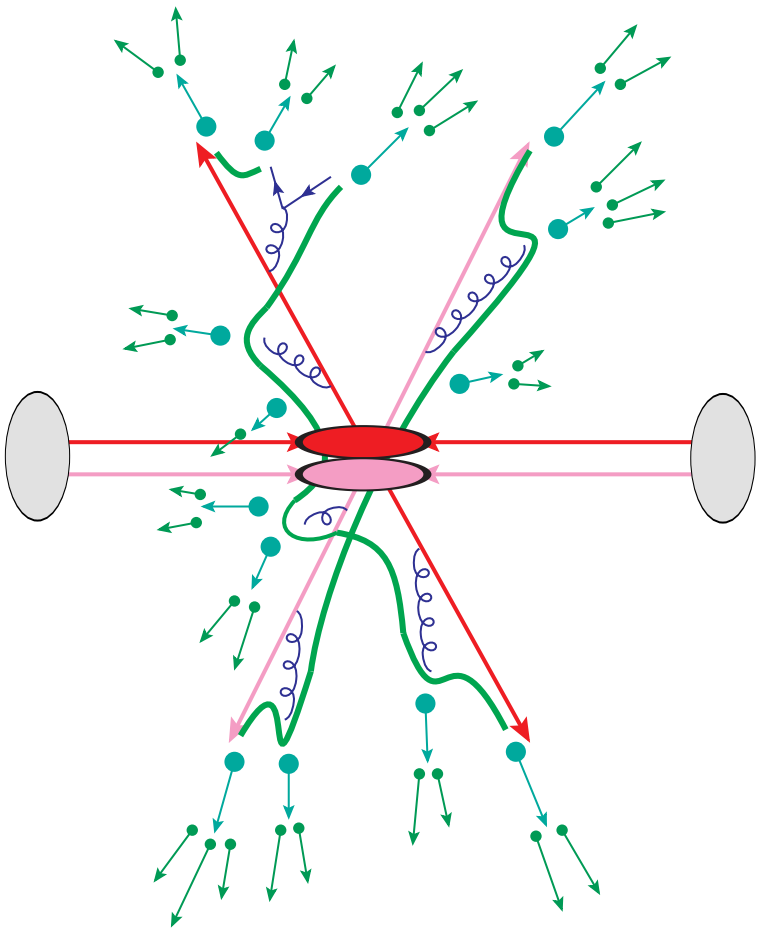
Hadronization Models in Generators

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Fermilab, CMS, Pythia

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What are Event Generators?

Machines that generate realistic, high-energy particle collision event records using all the Standard Model physics we know and inspired models to fill in the rest.



Many diverse phenomena

Short-distance cross section:
μ_r^H, μ_f^H, PDF^H, α_s^H
Parton shower:
μ_q^{PS}, μ_r^{PS}, μ_f^{PS}, μ_{cut}^{PS}, PDF^{PS}, α_s^{PS}
Multiple interactions:
μ_q^{MPI}, PDF^{MPI}, α_s^{MPI} ...
String fragmentation:
$f(z)$, string p_T, tension
Beams:
Primordial k_T, remnants
Particle Decays:
BRs, MEs, BE correlations
...

Testing the SM requires more than HO perturbation theory and N^{m}LL parton showers

Why Hadronization Models?

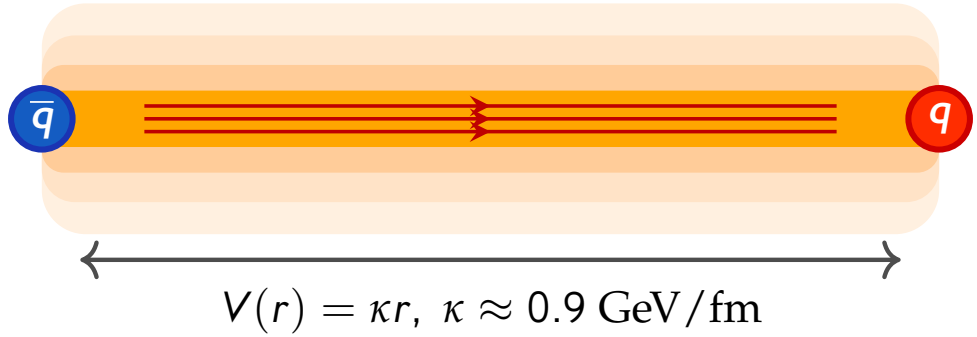
One of the most important yet least studied aspect of event generation

- Because we want to connect messy data to perturbative SM
- Hadronization is "just" a nuisance
- Theory has not produced quantitative, particle-level predictions
- Now, many tests of SM limited by models
- Uncertainty that won't just go away

- Charge: confront Hadronization head-on, not as a nuisance, but as interesting science

The Lund String Model: Key component

QCD@large charge separation: field lines seem to be compressed to tubelike region(s) $\Rightarrow$ **string(s)**



Linear confinement with string tension:

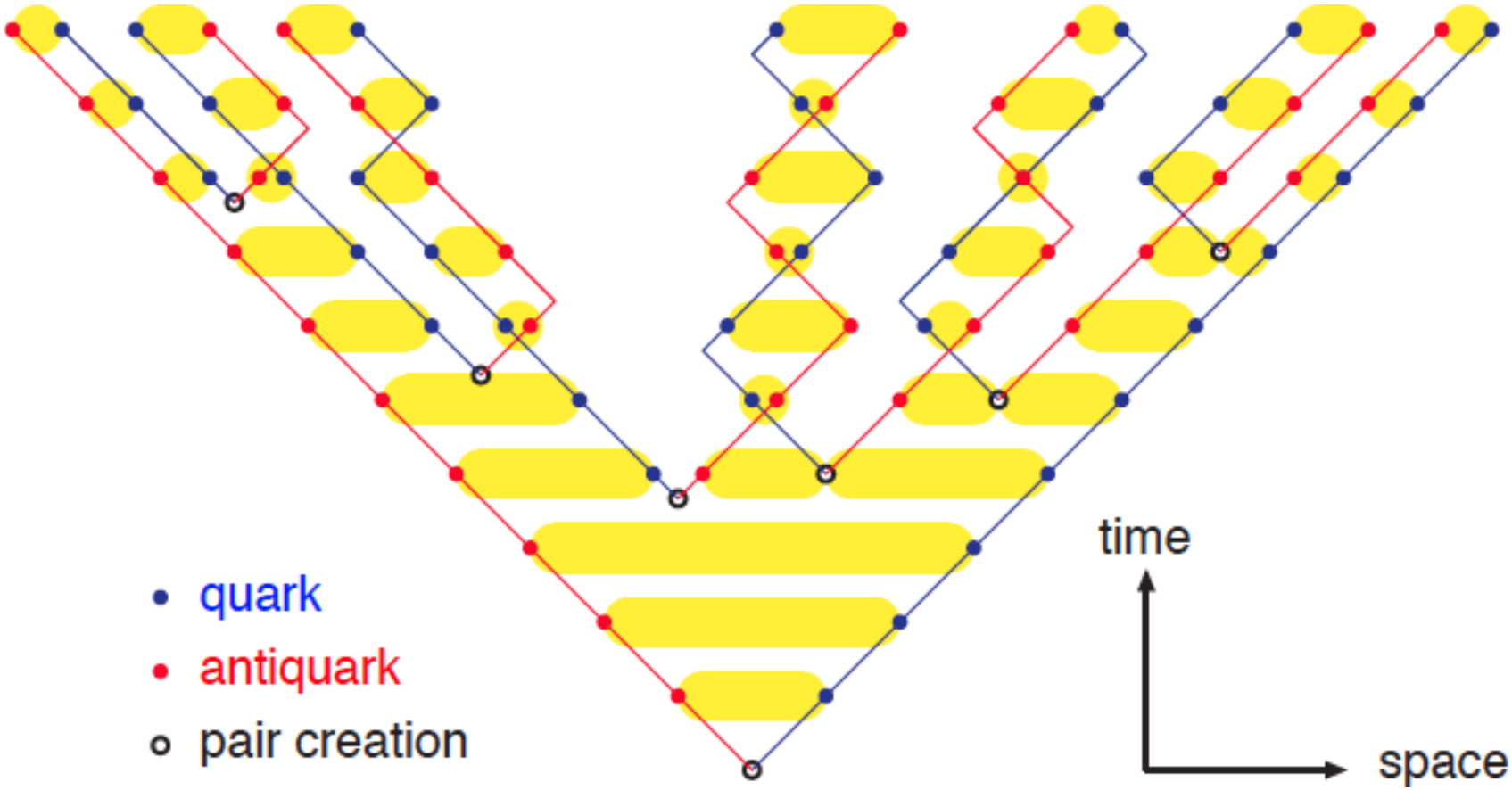
$F(r) \approx \mathrm{const} = \kappa \approx 1\,\mathrm{GeV/fm} \iff V(r) \approx \kappa r$

String breaks into hadrons along its length,
with roughly uniform probability in rapidity,
by formation of new $\mathrm{q\bar{q}}$ pairs that screen endpoint colors.

Simple 1+1-d object that is Lorentz covariant

Lund String Model: 2 jets

$e^+e^- \to q\bar{q}$



Consistent description of 2 jet topologies

String breakup viewed as "standard" ME calculation

$$d\mathcal{P} = |M|^2 \times dLIPS$$
$$= e^{-bA} \times \delta^{(2)}(\sum_i p_i - P_{\text{tot}}) \prod_{i=1}^{n} N\, d^2 p_i \delta(p_i^2 - m_i^2)\theta(E_i)$$

ME: Wilson area law for a confining field

$m_i \to m_{\perp_i}$: physical hadron mass plus some $p_\perp$

$N \to a$: related to Regge intercept

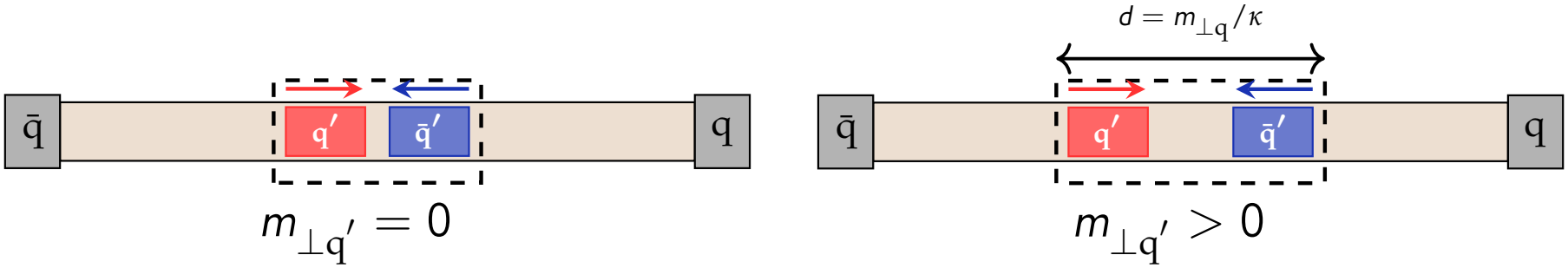
Bowler: massive quarks have reduced string area

Lund model uses iterative procedure to solve break-up:

$$p^+ p^- = m^2, p^+ = zP^+ \to p^- = \frac{m^2}{zP^+}$$

How Does the String Break?

Schwinger tunneling in a constant color-electric field



String breaking modeled by Schwinger tunneling:

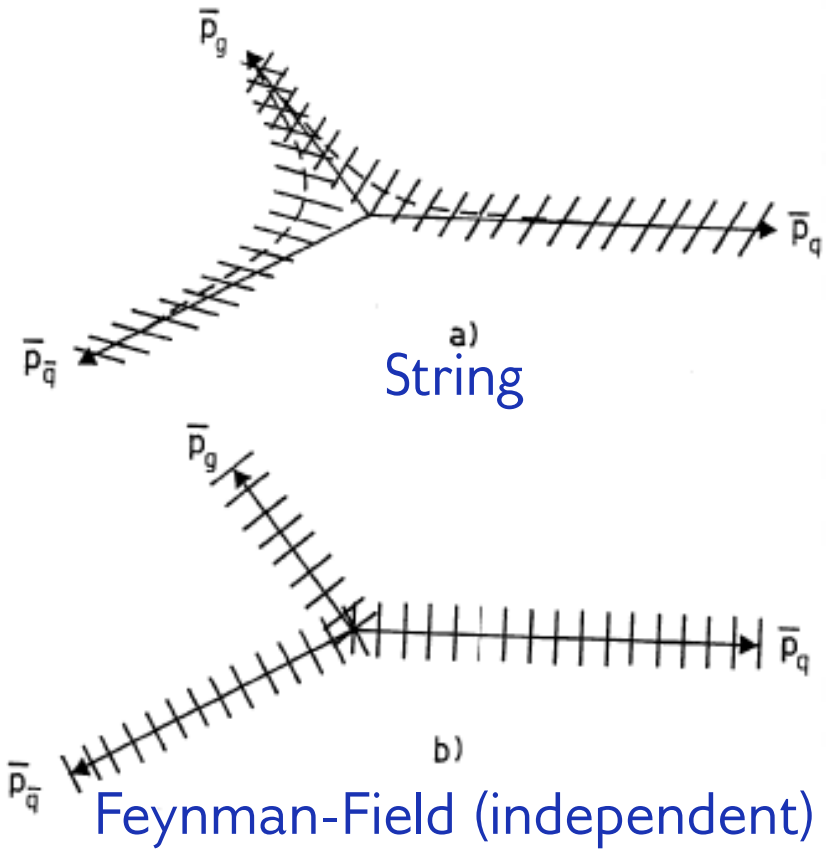
$$\mathcal{P} \propto \exp\left(-\frac{\pi m_\perp^2}{\kappa}\right) = \exp\left(-\frac{\pi p_{\perp q}^2}{\kappa}\right) \exp\left(-\frac{\pi m_q^2}{\kappa}\right)$$

- Common Gaussian $p_\perp$ spectrum, $\langle p_\perp \rangle \approx 0.3$ GeV
  Diluted by boost from local rest frame of string
- Suppression of heavy quarks:
  $u\bar{u} : d\bar{d} : s\bar{s} : c\bar{c} \approx 1 : 1 : 0.3 : 10^{-11}$
- Diquark $-$ antiquark $\Rightarrow$ simple model for baryon production

Tunneling already studied on quantum hardware.

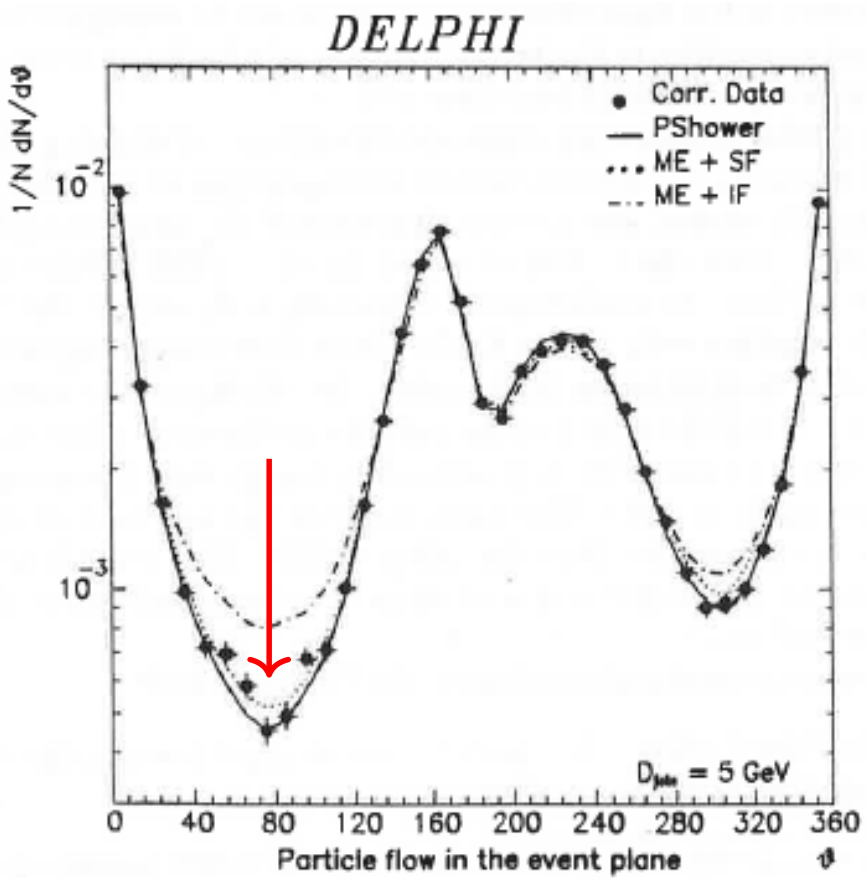
Lund String Model: 3 jets

No new parameters introduced for gluon jets!



String

Feynman-Field (independent)



Numerous and more detailed tests at LEP and other colliders

The Cluster Model (Wolfram, Webber)

Alternative to string fragmentation — used in Herwig/Sherpa

"Preconfinement": adjacent partons in a shower form low-mass color-singlet clusters at scale Q_0.

HERWIG basic scheme:

1. Force $g \to q\bar{q}$ branchings ($m_g > 2m_{u/d}$, from LQCD)
2. Form color-singlet clusters
3. Decay high-mass clusters to smaller clusters along "string" direction
4. Decay clusters to 2 hadrons (phase space $\times$ spin weight)

Cluster breaks *and* coherent shower branchings trace string topology $\Rightarrow$ similar particle flow patterns to Lund model.

Fewer parameters than string model, but less flexibility for heavy flavor.

How to study Hadronization?

- TEST MODELS!
  Many parameters
  Simulated using MC methods
- How to streamline? Reweighting
  Bookkeeping of each acceptance and rejection in MC chain
  kinematic AND flavor reweighting
- GOAL: Make PYTHIA as reweightable (differentiable) as possible
- NEW: *in-situ* (as in parton shower) AND *post-hoc* hadronization variations

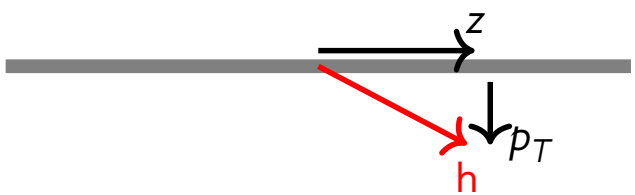
How many parameters? Next slide:

String-breaking parameters.

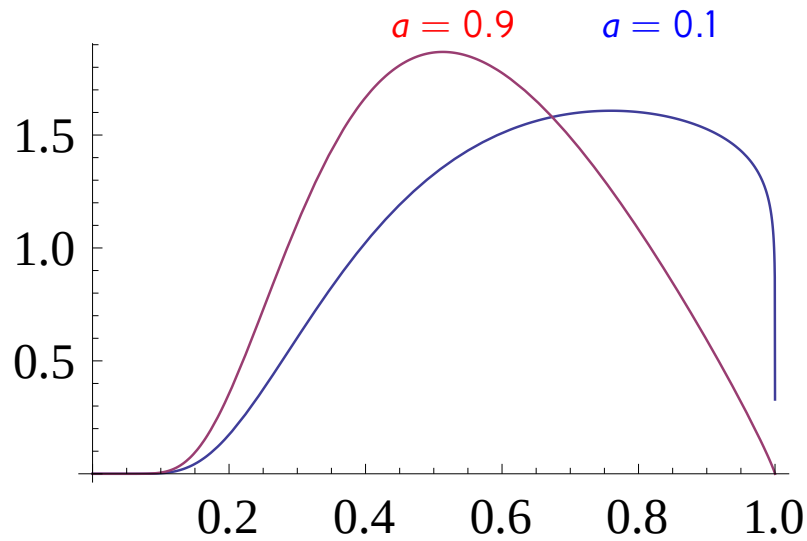
HAD Parameters	Monash 13	(Default)	Comment
# String breaks: pT and z distributions			
StringPT:sigma	= 0.335	= 0.304	! Soft pT in string breaks (in GeV)
StringPT:enhancedFraction	= 0.01	= 0.01	! Fraction of breakups with enhanced pT
StringPT:enhancedWidth	= 2.0	= 2.0	! Enhancement factor
StringZ:aLund	= 0.68	= 0.3	! Lund FF a (hard fragmentation supp)
StringZ:bLund	= 0.98	= 0.8	! Lund FF b (soft fragmentation supp)
StringZ:aExtraSquark	= 0.0	= 0.0	! Extra a when picking up an s quark
StringZ:aExtraDiquark	= 0.97	= 0.50	! Extra a when picking up a diquark
StringZ:rFactC	= 1.32	= 1.00	! Lund-Bowler c-quark parameter
StringZ:rFactB	= 0.855	= 0.67	! Lund-Bowler b-quark parameter
# Flavour composition: mesons			
StringFlav:ProbStoUD	= 0.217	= 0.19	! Strangeness-to-UD ratio
StringFlav:mesonUDvector	= 0.5	= 0.62	! Light-flavour vector suppression
StringFlav:mesonSvector	= 0.55	= 0.725	! Strange vector suppression
StringFlav:mesonCvector	= 0.88	= 1.06	! Charm vector suppression
StringFlav:mesonBvector	= 2.2	= 3.0	! Bottom vector suppression
StringFlav:etaSup	= 0.60	= 0.63	! Suppression of eta mesons
StringFlav:etaPrimeSup	= 0.12	= 0.12	! Suppression of eta' mesons
# Flavour composition: baryons			
StringFlav:probQQtoQ	= 0.081	= 0.09	! Diquark rate (for baryon production)
StringFlav:probSQtoQQ	= 0.915	= 1.000	! Strange-diquark suppression
StringFlav:probQQ1toQQ0	= 0.0275	= 0.027	! Vector diquark suppression
StringFlav:decupletSup	= 1.0	= 1.0	! Spin-3/2 baryon suppression
StringFlav:suppressLeadingB	= off	= off	! Optional leading-baryon suppression
StringFlav:popcornSpair	= 0.9	= 0.5	!
StringFlav:popcornSmeson	= 0.5	= 0.5	!

Lund symmetric fragmentation function:

$$f(z) \propto \frac{(1-z)^a}{z} \exp\left(\frac{-bm_\perp^2}{z}\right)$$

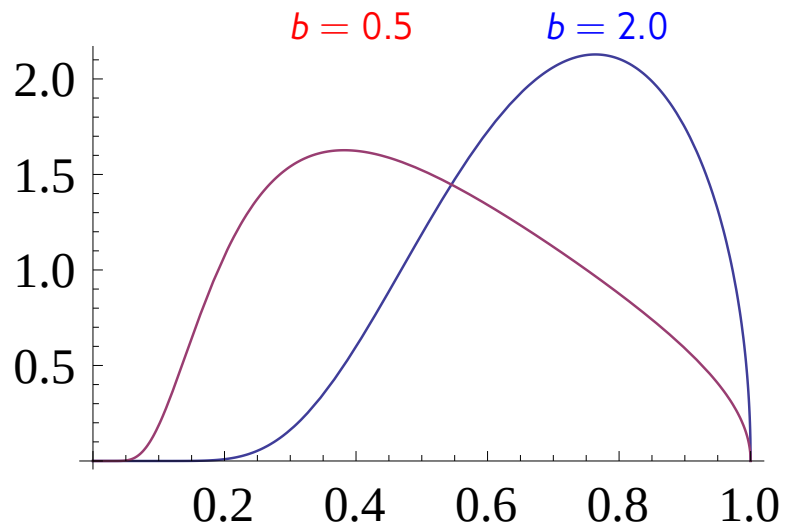


The a parameter

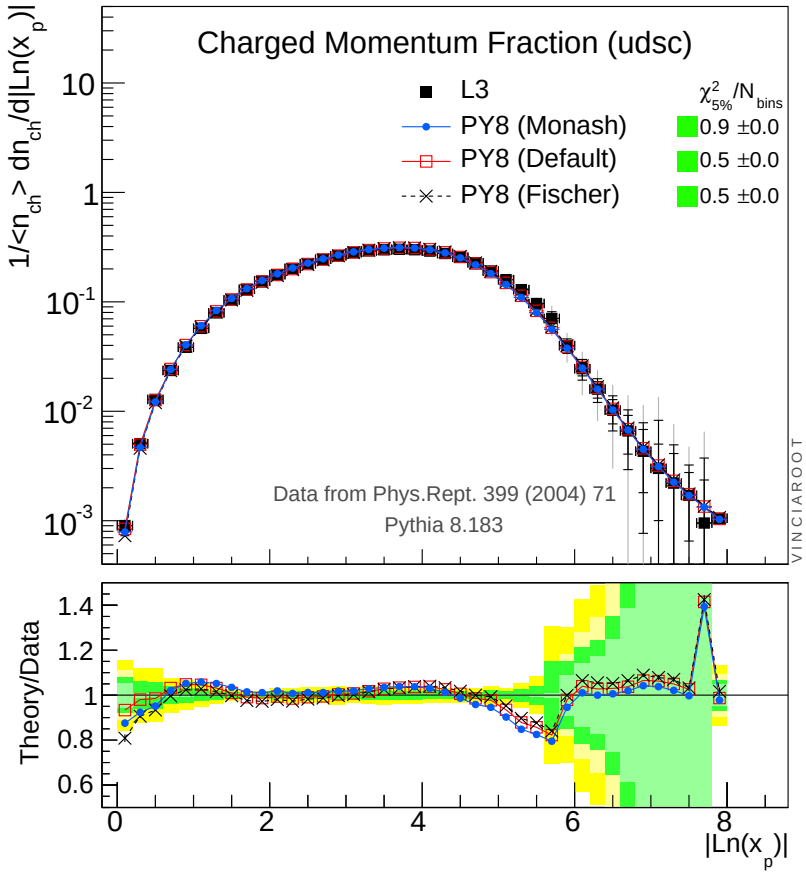


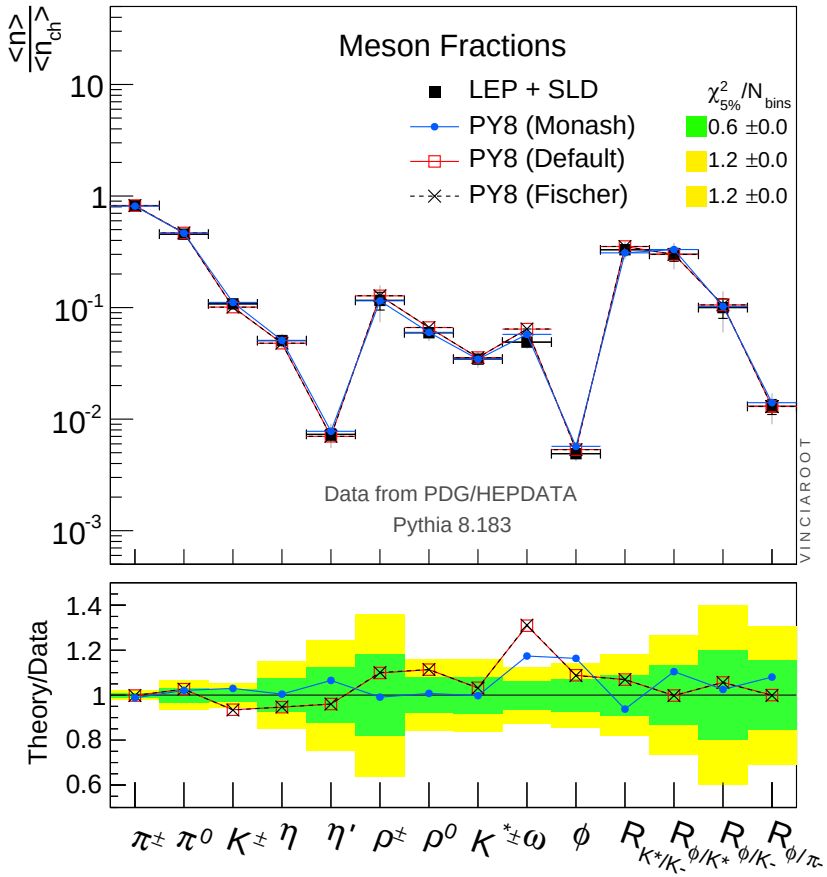
$b = 1\text{GeV}^{-2}$, $m_\perp = 1\text{GeV}$

The b parameter



$a = 0.5$, $m_\perp = 1\text{GeV}$



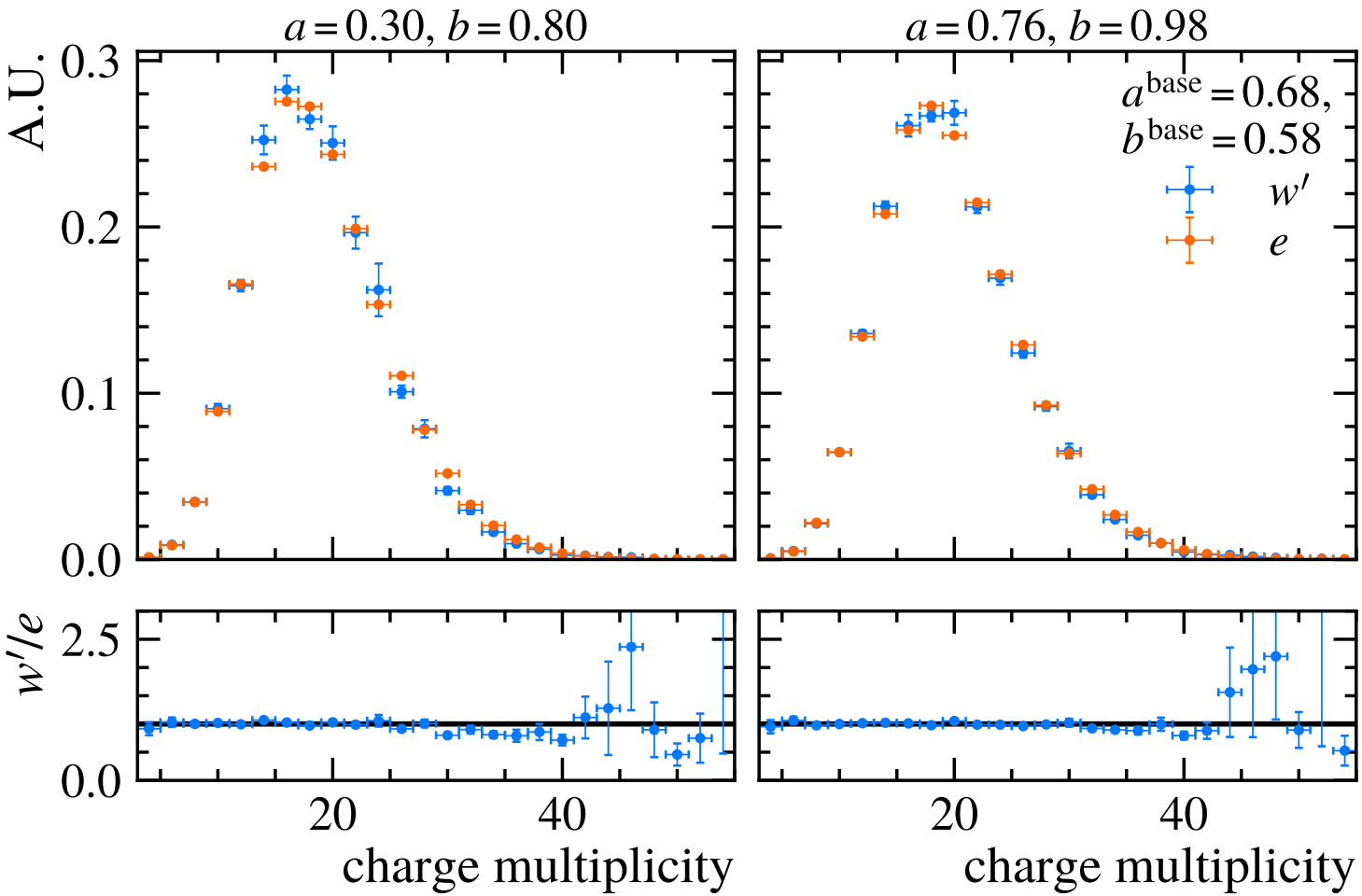



```
StringZ:aLund          =  0.68
StringZ:bLund          =  0.98

StringZ:aExtraDiquark =  0.97
StringZ:aExtraSquark  =  0.00
```

```
StringFlav:ProbStoUD     = 0.217
StringFlav:mesonUDvector = 0.5
StringFlav:mesonSvector  = 0.55
StringFlav:etaSup        = 0.60
StringFlav:etaPrimeSup   = 0.12
```

String splitting variations



Reweighting of baseline calculation to different values of a, b versus an explicit calculation.

Flavor variations

$\rho \equiv$ `probStoUD`

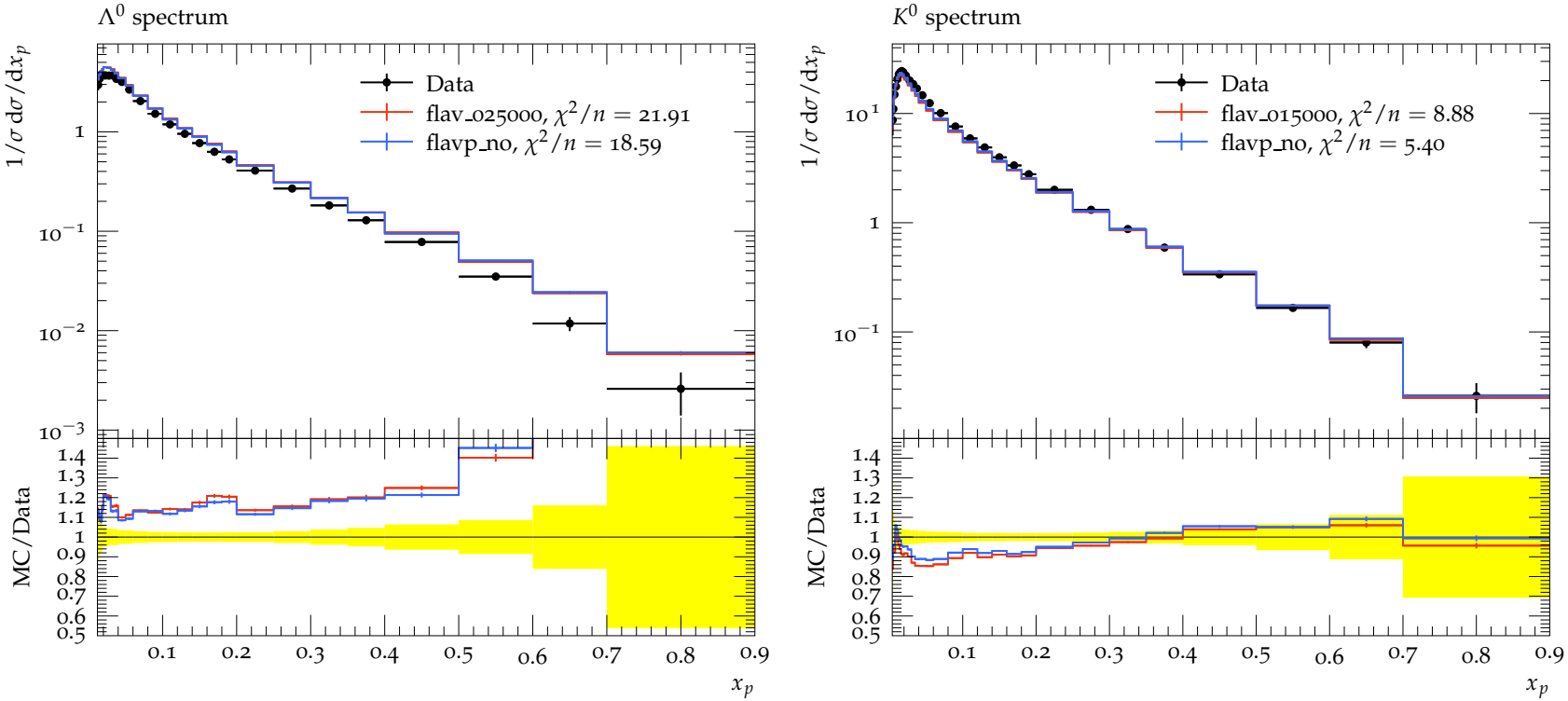


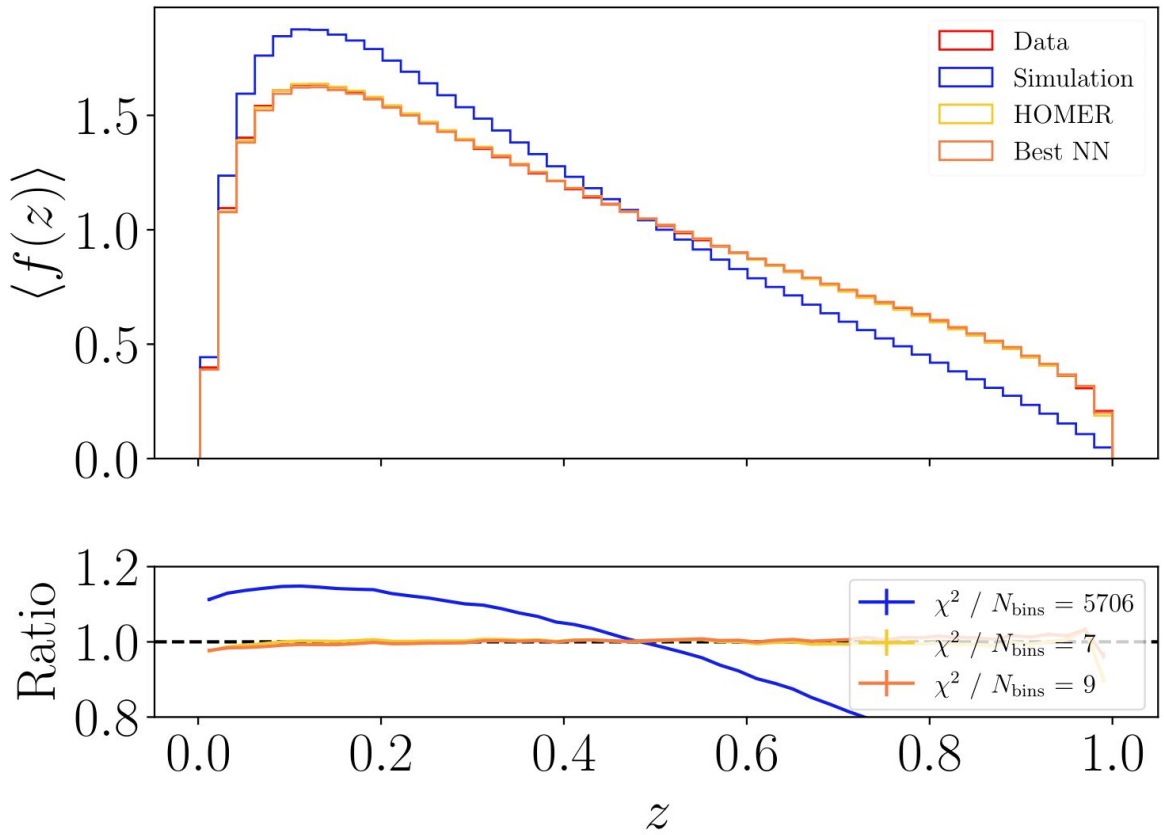
Figure: Λ^0 energy fraction reweighting $\rho = 0.217 \rightarrow \rho = 0.25$ (l) and K^0 energy fraction reweighting $\rho = 0.217 \rightarrow \rho = 0.15$ (r). Popcorn splitting has been disabled.

Hadronization is Machine Learnable

- Models are complex:
  Why not use complex parametrization (ML)?
- Current models "work" and are "understandable."
- ML-**enhanced** string model: ML a correction
- Reweighting an asset:
  "easy" to explore parameter space, train, validate

Data-driven hadronization model

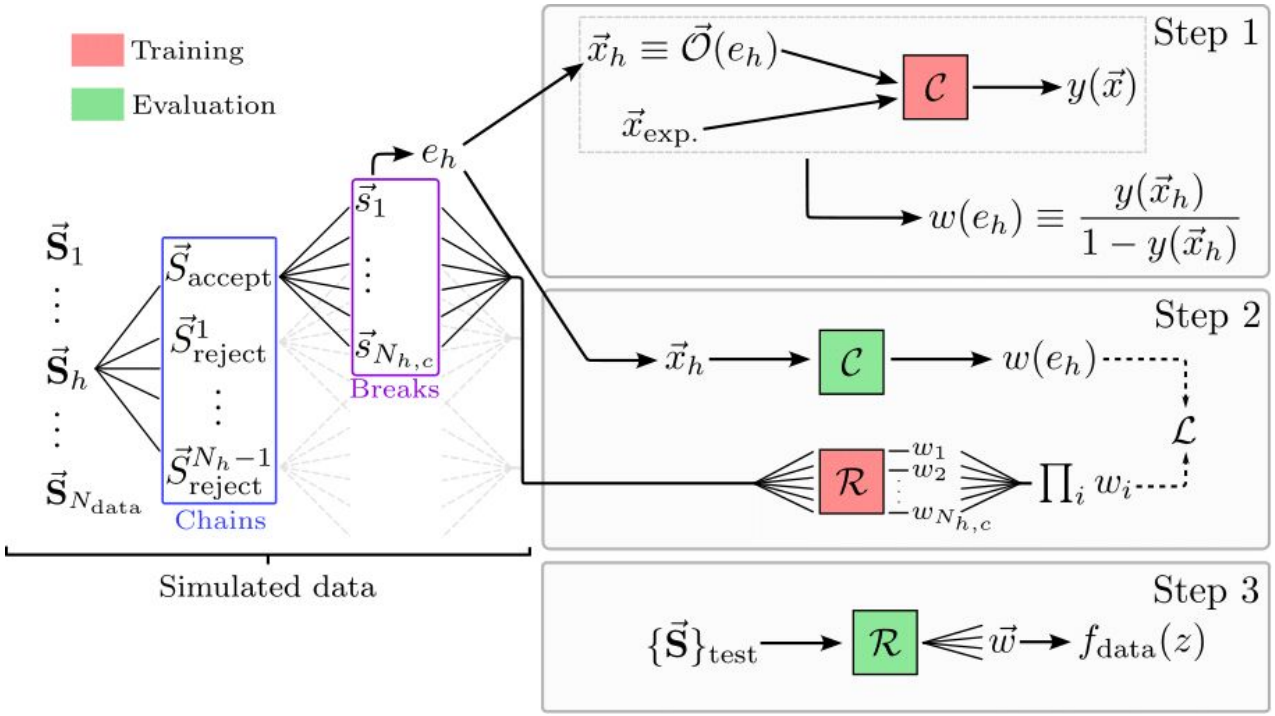
Correcting the Lund fragmentation function(a,b)



$$f(z) \propto \frac{(1-z)^a}{z} \exp\left(-\frac{bm_{\mathrm{T}}^2}{z}\right)$$

HOMER: Histories and Observables for Monte carlo Event Reweighting

HOMER Workflow



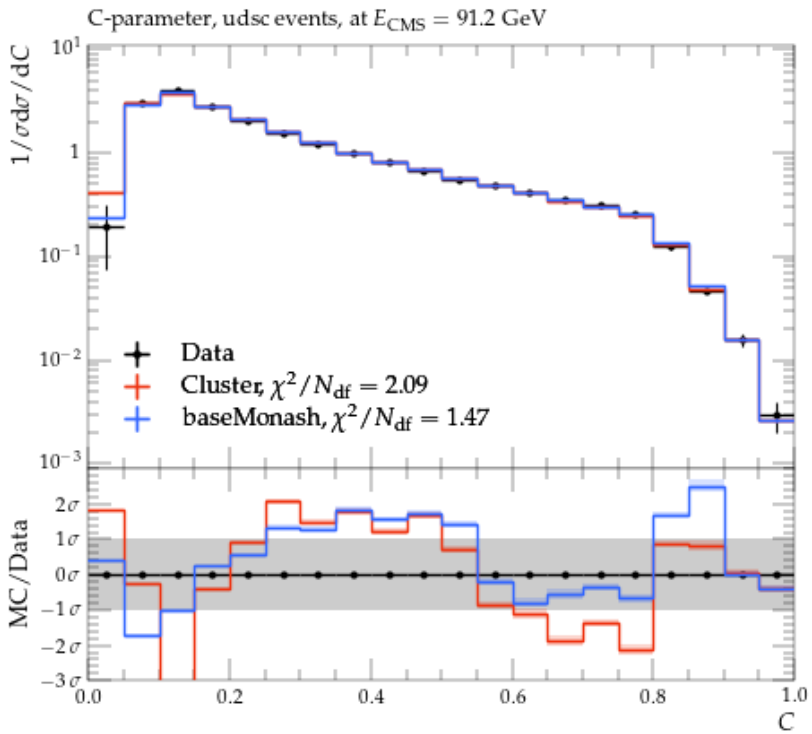
Generate ONCE and learn reweighting function:
New Model is Pythia + weights

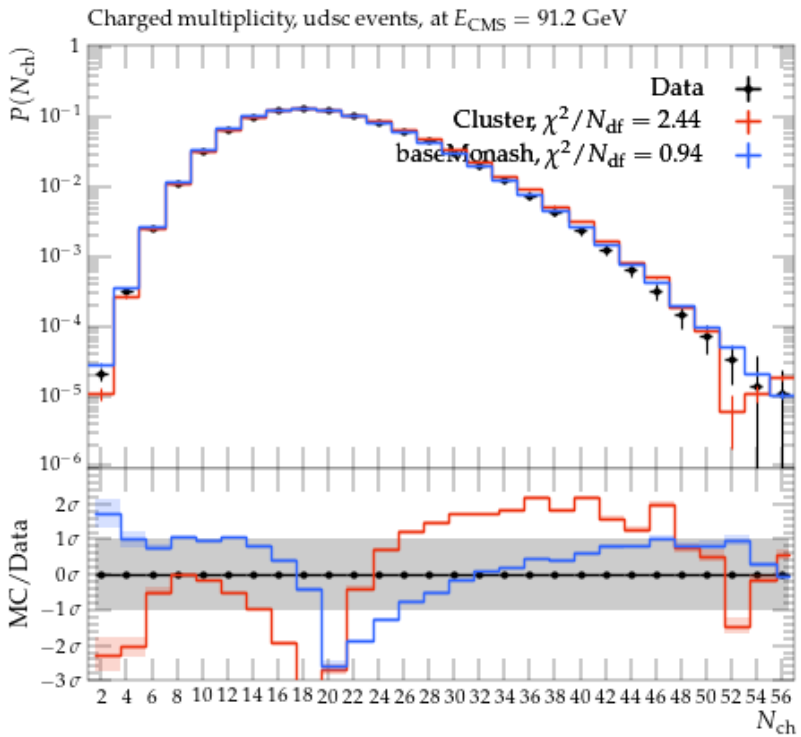
New plugin capability in PYTHIA

Example: SHERPA CLUSTER hadronization model with PYTHIA shower

```
Jet rates and event shapes at LEP I+II (L3_2004_I652683)

| HepData
| 10.1016/j.physrep.2004.07.002
```





Extensions of the String Model

- Rope hadronization (Bierlich et al.): overlapping strings in dense environments $\rightarrow$ enhanced strangeness $\rightarrow$ collective effects resembling quark–gluon plasma
- String shoving: repulsion between color strings, azimuthal anisotropy without hydrodynamics
- Thermal string model: add thermal Boltzmann factor to tunneling probability, temperature T_{eff}
- Beyond Gaussian $p_{\perp}$: non-Gaussian tails, curved string segments
- Baryon junctions: baryons from string junction rather than diquark; important for baryon number transport

All extensions introduce more parameters.
Lattice or quantum simulation input would be valuable.

Some take-away points

- Phenomenological Hadronization models look complicated
  Lund string is 1+1D model with Gaussian p_T
- Somewhat factorizable from rest of simulation
  Missing perturbative physics will be absorbed, though
- Reweightable (implemented in Pythia)
- Machine Learnable (see HOMER)

Desired outcome from this workshop

- Identify simple 1+1D models that exhibit properties of confinement and string-breaking that can be studied on quantum computers/simulators and turned into particle-level predictions

$$H_2 = \tau \sum_{j_1<j_2} (j_2 - j_1) \, |j_1, j_2\rangle\langle j_1, j_2|$$
$$+ \omega \sum_{j_1<j_2} \Big[|j_1 - 1, j_2\rangle\langle j_1, j_2| + |j_1 + 1, j_2\rangle\langle j_1, j_2| + \text{h.c.} \Big] + \text{pair-creation}$$

j_1, j_2: lattice sites; ω: hopping; τ: string tension
(Surace, Lerose)

- Stop trying to solve the hard problem first!